

VIDEO_COURSE_CURRICULUM

HelixCode Video Course Curriculum

Status: Planned **Format:** Screen-capture with voiceover + Code-along exercises.

Course 1: HelixCode Fundamentals (Beginner)

Goal: Get users from zero to their first running agent.

Module 1: Introduction & Setup

- **Video 1.1:** What is HelixCode? (Architecture Overview).
- **Video 1.2:** Installation (Linux, Mac, Windows, Docker).
- **Video 1.3:** Initial Configuration (Setting up API Keys).
- **Video 1.4:** The CLI & TUI: A Tour.

Module 2: Core Concepts

- **Video 2.1:** Understanding Providers (Ollama vs. OpenAI vs. Claude).
- **Video 2.2:** Managing Sessions (Plan, Build, Test, Deploy).
- **Video 2.3:** Working with Tasks & Checkpoints.
- **Video 2.4:** The Worker Pool: Distributed Computing Explained.

Module 3: Your First Project

- **Video 3.1:** Project Scaffolding.
- **Video 3.2:** Generating Code with Templates.
- **Video 3.3:** Running Tests & Debugging.

Course 2: Advanced Agentic Workflows (Intermediate)

Goal: Master the tool system and complex workflows.

Module 4: The Tool Ecosystem

- **Video 4.1:** File System Tools (Safe Editing).
- **Video 4.2:** Web Tools (Search & Fetch).
- **Video 4.3:** Shell Execution & Sandboxing.

- **Video 4.4:** Multi-File Atomic Edits.

Module 5: Memory & Context

- **Video 5.1:** Short-term vs. Long-term Memory.
- **Video 5.2:** Configuring Vector Databases (Chroma, Weaviate).
- **Video 5.3:** Context Building Strategies.
- **Video 5.4:** Using “Extended Thinking” Models.

Module 6: Custom Workflows

- **Video 6.1:** Defining Custom Workflows in YAML.
- **Video 6.2:** Chaining Agents.
- **Video 6.3:** Human-in-the-Loop Workflows.

Course 3: Enterprise Architecture (Advanced)

Goal: Deploy HelixCode in secure, large-scale environments.

Module 7: Enterprise Providers

- **Video 7.1:** Connecting to AWS Bedrock.
- **Video 7.2:** Connecting to Azure OpenAI.
- **Video 7.3:** Connecting to Google Vertex AI.
- **Video 7.4:** Managing Rate Limits & Costs.

Module 8: Security & Deployment

- **Video 8.1:** Authentication & RBAC.
- **Video 8.2:** Secure Worker Deployment (SSH Keys, Firewalls).
- **Video 8.3:** Monitoring with Prometheus & Grafana.
- **Video 8.4:** Docker Swarm / Kubernetes Deployment.

Course 4: Extending HelixCode (Expert)

Goal: Contribute to the core or build plugins.

Module 9: Plugin Development

- **Video 9.1:** The MCP Protocol Deep Dive.
- **Video 9.2:** Building a Custom Tool.
- **Video 9.3:** Building a Custom Provider.

Module 10: Contribution Guide

- **Video 10.1:** Codebase Walkthrough.
- **Video 10.2:** Running the Test Suite.
- **Video 10.3:** Submitting a PR.